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Amrita School of Computing, Amritapuri Campus

Department of Artificial Intelligence and Data Science

23AID204

Advanced Data structures and Algorithm Analysis

L-T-P-C:3-0-2-4

Lab Assignment 3

Binary search tree

Course Outcomes: CO3

[BTL 3]

Objectives

After completing this lab, students will be able to:

- To implement and perform basic Binary Search Tree operations such as insertion, deletion, searching, and traversal.
- To apply BST concepts for solving simple real world and application oriented problems.

In - Lab Assignment

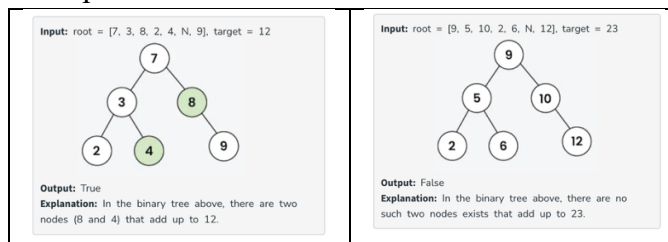
1. A department wants to maintain student roll numbers using a Binary Search Tree.

Perform the following operations sequentially:

- Insert the roll numbers: 45, 30, 70, 20, 40, 60, 80, 35, 50
- Search for roll number 35 and 90.
- Delete roll number 30.
- Display the tree using Inorder, Preorder and Postorder
- Explain why inorder traversal gives roll numbers in sorted order.

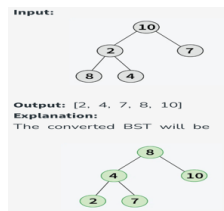
2. Given a Binary Search Tree(BST) and a target. Check whether there's a pair of Nodes in the BST with value summing up to the target.

Example:



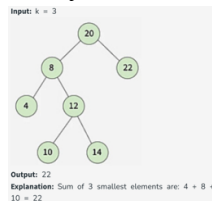
3. Given a Binary Tree, convert it to Binary Search Tree in such a way that keeps the original structure of Binary Tree intact.

Example:



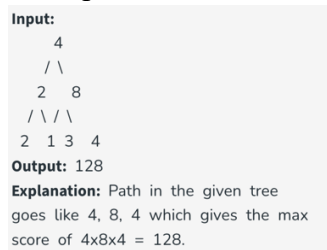
Take - home Assignment

- Given a Binary Search Tree, find sum of k smallest nodes in it. You may assume that k is always smaller than or equal to size of the tree.



- In an online game, N blocks are arranged in a hierarchical manner. All the blocks are connected together by a total of N-1 connections. Each block is given an ID from 1 to N. A block may be further connected to other blocks. Each block is also assigned a specific point value. A player starts from Block 1. She must move forward from each block to a directly connected block until she reaches the final block, which has no further connection. When the player lands on a block, she collects the point value of that block. The players score is calculated as the product of point values that the player collects. Find the maximum score a player can achieve.

Example:



- Given a Binary Search Tree, the task is to count the number of nodes which are having special two-digit numbers.

Note: Special two-digit numbers are those numbers that satisfy the condition that the sum of their digits added to the product of their digits equals the original number.

Example:

